



AI-Powered Collections Strategy



Leveraging Agentic AI for Scalable, Fair, and
Effective Debt Management at Geldium



How the System Works

- **Inputs:**
Real-time customer data (e.g., income, credit utilization, missed payments, payment history)
- **Decision Logic:**
Combines predictive scores and business rules to determine optimal actions.
Decision engine applies the rules to create targeted interventions
- **Actions:**
Automated outreach: payment reminders, hardship offers, repayment plans.
Tailored customer engagement through SMS, email, or phone calls
- **Learning:**
Continuously refines the system's actions based on real-time feedback (e.g., repayment rates, customer engagement)

System Components and Workflow

- Data Pipeline
 - Gathers essential customer information, including financial behavior (e.g., income, missed payments, credit utilization)
- Decision Engine
 - Applies machine learning models and business rules to determine actions
 - Factors in customer's risk level, payment history, and predicted behavior
- Action Layer
 - Executes interventions: automated SMS/email reminders, customized repayment offers, hardship assistance
- Learning Loop
 - System adapts based on the results of previous actions
 - Future decision-making improves based on repayment outcomes

Role of Agentic AI

Agentic AI autonomously handles routine collections tasks, while human oversight ensures critical decisions are fair and nuanced

Autonomous	Human Oversight
Sending automated reminders (SMS/email)	Offering tailored debt restructuring plans
Providing general repayment offers	Reviewing escalations or legal actions
Follow-up for low-risk customers	Verifying high-risk cases (e.g., legal action)
Adaptive behavior based on customer actions	Manual review of sensitive or complex scenarios

Responsible AI Guardrails

Fairness

- Regular bias audits to ensure equal treatment of all customer segments
- Minimize disparate impact based on demographics or financial status
- Transparency
 - Clear, accessible explanations for all AI-driven decisions
 - Customers are informed of their rights and the basis of decisions
- Compliance
 - System aligns with GDPR, ECOA, and relevant financial regulations (e.g., consumer protection laws)
- Oversight
 - Human intervention required for critical decisions or sensitive cases (e.g., denying hardship assistance)

Expected Business Impact

- Business Outcomes
 - 15% reduction in 30+ day delinquency for high-risk customers within the first 6 months.
 - Automate 60% of outreach actions, reducing operational costs
- Customer Outcomes
 - Improve customer satisfaction through timely, respectful, and tailored interventions.
 - Enhance trust by offering transparent, fair, and empathetic outreach
- Operational Efficiency
 - Scale operations by automating routine tasks, allowing staff to focus on more complex cases

Impact on Geldium's Collections Strategy

How This AI-Powered System Benefits Geldium

- Enhanced Scalability: AI-driven automation allows for large-scale outreach without compromising personalized service
- Cost Efficiency: Significant reduction in manual outreach, reducing labor costs
- Improved Risk Management: Predictive models help target interventions to at-risk customers, improving collection efficiency
- Better Customer Experience: Fair and transparent system increases customer trust and satisfaction, leading to higher repayment rates